Heaven's Light Is Our Guide
Rajshahi University of Engineering & Technology
Department of Electrical & Electronic Engineering

Proposal for Capstone Project Design I (EEE 3202)

1.	**Project member**	:	Labib Marwan Hoque (2101064)
2.	**Title of the project**	:	Real-Time Edge-Based Facial Recognition System Using Deep Learning on Raspberry Pi 4

3. Brief description of the project:

(a) Project Background:

- **Problem Statement:**
  Biometric security and smart automation are rapidly becoming essential components of modern infrastructure. However, most existing facial recognition systems rely heavily on **Cloud Computing**, which introduces significant issues regarding **data privacy** (sending biometric data to external servers), **network latency** (lag due to internet speed), and high operational costs. Conversely, traditional edge-based solutions (running locally on devices) face a severe hardware bottleneck. Algorithms like **Haar Cascades** are fast but suffer from poor accuracy in varying lighting or side profiles, while modern Convolutional Neural Networks (CNNs) like **Dlib/HOG** are accurate but computationally too expensive for cost-effective hardware like the Raspberry Pi, often resulting in unusable frame rates (below 1 FPS).
- **Existing Research & Gap:**
  Current literature explores lightweight models like MobileNet or SqueezeNet, yet implementation on general-purpose single-board computers (SBCs) often requires specialized accelerators (like Google Coral or Nvidia Jetson). There is a distinct lack of optimized software pipelines that can leverage the native CPU of a standard Raspberry Pi 4 to perform **Deep Learning inference** in real-time without extra hardware.
- **My Approach:**
  This project addresses this gap by implementing a **Quantized Deep Learning pipeline** using OpenCV's DNN module. We utilize **YuNet** (a quantized nanometer-scale detector) for face detection and **SFace** (Sigmoid-Constrained Hypersphere Loss) for recognition. This combination allows us to bypass the computational heaviness of traditional methods, enabling a secure, offline, and real-time biometric system that runs entirely on the edge.

(b) Objectives and aims of the project:

The primary objective is to develop a standalone, offline facial recognition system. Specific aims include:

- **Objective 1:** To implement an optimized Deep Learning pipeline (YuNet + SFace) on the Raspberry Pi 4.
- **Objective 2:** To achieve a usable real-time frame rate of **3–5 FPS** without external hardware accelerators.
- **Objective 3:** To develop a robust "One-Shot" encoding system that allows new users to be registered instantly without re-training the model.
- **Objective 4:** To implement dynamic tracking of unregistered individuals ("Unknown" faces) for security monitoring.

(c) State how your project address the following program outcomes (PO):

POs for EEE 3202	PO Statements
PO(e) (Modern tool usage)	The project extensively utilizes modern software engineering and computer vision tools. We are applying **Python 3.11** for logic, **OpenCV 4.6.0 (DNN module)** for AI inference, and **Picamera2** for low-latency hardware control. Furthermore, we use **ONNX (Open Neural Network Exchange)** models for platform-agnostic deep learning deployment, demonstrating the application of advanced IT tools to solve complex edge-computing limitations.
PO(i) (Individual work and teamwork)	As a member/leader of this project, I am responsible for the end-to-end system lifecycle. This involves individual research into model quantization, configuring the Linux environment (Raspberry Pi OS Bookworm), and system integration. If working in a team, roles are divided between hardware setup (cooling/wiring) and software optimization, ensuring effective collaboration in a multidisciplinary setting combining electronics and computer science.
PO(j) (Communi-cation)	The project involves generating comprehensive design documentation, including block diagrams of the recognition pipeline and performance analysis charts (FPS vs. Accuracy). The system itself communicates visually with the user through a designed UI (Green/Red bounding boxes and confidence scores), and the final outcomes will be presented through a technical poster and report.
PO(k) (Project manageme- nt and finance)	The project is managed within a strict budget constraint, prioritizing cost-effectiveness by using off-the-shelf components (Pi 4, Standard Camera) rather than expensive industrial sensors. Time management is applied to ensure the software development lifecycle (Design →Train →Test) is completed within the academic timeline.

(d) Methodology to be adopted in the project:

The project follows a linear processing pipeline:

1. **Hardware Setup:** A Raspberry Pi 4 (4GB) is coupled with a 5MP OV5647 Camera Module via the CSI interface. Active cooling is applied to prevent thermal throttling during AI inference.
2. **Image Acquisition:** Video is captured at 640×480 resolution using Picamera2 in RGB888 format to ensure color accuracy for the AI models.
3. **Pre-processing:** Frames are mirrored (flipped) for user intuition and resized to 320×320 to match the input tensor requirements of the neural network.
4. **Inference Engine:**
   - **Detection:** The frame is passed to **YuNet**, which outputs bounding boxes and 5 facial landmarks.
   - **Recognition:** Detected faces are aligned and passed to **SFace**, which generates a 128-dimensional embedding vector.
5. **Matching Logic:** The system calculates **Cosine Similarity** between live embeddings and a stored .pickle database. A threshold of **0.363** determines if a face is "Known" or "Unknown".

(e) Expected results:

- **Performance:** A stable frame rate of **3–5 FPS** on the CPU, which is sufficient for access control scenarios (e.g., walking up to a door).
- **Accuracy:** Greater than **98% accuracy** in identifying registered users, with robust performance in low-light and side-profile views (up to ±30∘±30∘).

- **Functionality:** Successful distinction between multiple registered users and tracking of unregistered intruders simultaneously.

(f) Relevance of the project to national development:

This project supports the **"Digital Bangladesh"** and **"Smart Bangladesh 2041"** initiatives by providing a low-cost, domestically producible solution for security and digitization. It can be deployed for **automated attendance in schools**, **secure entry in government offices**, and **low-cost surveillance** for small businesses, reducing reliance on expensive imported security solutions.

4. Basic facilities available in the department for the proposed investigation:

- Computer Laboratories for coding and simulation.
- Basic electronics equipment (Power supplies, Monitors, Peripherals) required to interface with the Raspberry Pi.
- Library access for research papers on Deep Learning quantization.

5. Key performance indicators of the project:

1. **Frames Per Second (FPS):** Target ≥ 3 FPS
2. **Inference Latency:** Target < 300ms per frame.
3. **True Positive Rate (Accuracy):** Target > 95%.
4. **False Acceptance Rate (FAR):** Target < 1% (Critical for security).

6. Cost Estimation:

Sl. No.	Item Description		Cost	
	Name of the Item	Quantity	Unit Price (BDT)	Total Cost (BDT)
1.	Raspberry Pi 4 Model B (4GB RAM)	1	12,850	12,850
2.	Raspberry Pi Camera Module (5MP)	1	880	880
3.	Micro SD Card (32GB)	1	600	600
4.	Power Bank	1	2000	2000
5.	Cable (USB to USB-C)	1	150	150
6.	Aluminum Heat Sink Set with Adhesive	1	120	120
			Total Expense (Taka)	16,600

Project Member:	**Supervisor:**
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